

## CONFERENCE REPORT

# Summary of the Inaugural ReNU Hope Conference and Scientific Symposium, July 23–25, 2025, Long Island, New York

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## ABSTRACT

The inaugural ReNU Hope Conference and Scientific Symposium was held from July 23–25, 2025 in Long Island, New York. This historic conference brought together the researchers responsible for the groundbreaking discovery of RNU4-2/ReNU syndrome, families, scientists, clinicians, trainees, therapeutic developers, and industry leaders from around the world. The key themes that emerged included: (1) Early recognition and diagnosis of ReNU Syndrome, (2) optimizing clinical care for this complex condition, (3) the importance of the family experience, (4) a need to elucidate the underlying genetic mechanism, (5) a need for quality natural history data and validated endpoints, and (6) exploring approaches to therapeutic development. This summary provides a broad overview of the conference, highlights the key presentations and discussions, and delineates priorities moving forward.

## 1 | Introduction

ReNU/RNU4-2 syndrome (RENU, OMIM#620851) is a recently described neurodevelopmental condition caused by heterozygous pathogenic variants in *RNU4-2*, a 145 base pair non-coding RNA gene which encodes the U4 snRNA component of the major spliceosome. Using large scale genome-sequenced cohorts, genetic association analysis, and RNA sequencing technologies, two research groups independently discovered and delineated the syndrome in April/May 2024 (Chen et al. 2024; Greene et al. 2024). Key clinical features include developmental delays, intellectual disability, seizures, ophthalmological features, musculoskeletal features, and a distinct facial appearance (Valenzuela et al. 2024). Since then, over 300 individuals have been diagnosed and registered with the rapidly established

advocacy organization (ReNU Syndrome United). ReNU syndrome is now projected to be one of the leading monogenetic causes for intellectual disability, affecting approximately 1 in 30,000 individuals (Barbour et al. 2025), with an expected global population of around 100,000 (De Jonghe et al. 2025).

Upon receiving the RNU4-2 diagnosis in March/April 2024, an online Facebook group and website were initiated by parents to connect with one another, and engagement quickly grew as more began receiving the diagnosis around the world. Although the disorder was originally suggested to be named NEDHFA (neurodevelopmental disorder with hypotonia, brain anomalies, distinctive facies and absent language), the parent group collaborated with researchers to officially rename the syndrome ReNU, as a play on the RNU gene and

to symbolize “renewed” hope for the future. After months of discussions and advice to organize a patient advocacy organization and hold a date for an inaugural meet-up, a group of six parents officially formed ReNU Syndrome United, incorporated as a 501(c)3 nonprofit in October 2024. The organization's mission is to build community, raise awareness, and advance research for ReNU syndrome. The organization has rapidly scaled and ReNU Syndrome United associations are now up and running in several other countries, with leaders of all of the national associations meeting regularly and utilizing cohesive branding and messaging across the globe. One of the first major initiatives of ReNU Syndrome United was to hold an inaugural conference and scientific symposium to bring together the research community and the families who had been searching for this diagnosis for years, or even decades.

Held in July 23–25, 2025, in Long Island, New York, the inaugural ReNU Hope Conference and Scientific Symposium brought together an international delegation of the researchers responsible for the groundbreaking discovery of *RNU4-2*, families, scientists, clinicians, trainees, and therapeutic developers. The conference was co-organized by ReNU Syndrome United (the U.S. ReNU organization), Dr. Ian Krantz (Cohen Children's Medical Center/Northwell Health), and Kelsey Crocker, MS, CGC (Cohen Children's Medical Center/Northwell Health).

The family day conference took place on July 24, 2025 at the Long Island Marriott, 101 Doolittle Boulevard, Uniondale, NY 11553, and included approximately 220 attendees from across the world. This day started with Jessica Margrill, President and one of the founding Board members of ReNU Syndrome United, with an inspiring welcome speech about the impact this discovery had on families with ReNU syndrome and the steps leading up to the conference. She shared her own personal experience about her son's diagnostic odyssey, how the ReNU syndrome diagnosis led to starting an online community, finding supportive mentors, and ultimately banding together with incredible board members to launch ReNU Syndrome United, the first non-profit dedicated to this condition. Dr. Ian Krantz followed her presentation, acknowledging the ReNU Syndrome United organization for their extraordinary work in planning and executing the first ReNU Hope conference. Subsequent presentations were focused on the discovery of ReNU syndrome and ending the diagnostic odyssey, insights into symptoms and future clinical care, the very first Unique guide for ReNU syndrome, a ReNU sibling spotlight, a ReNU parent panel, navigating the therapeutic landscape, and a call for advocacy.

The scientific symposium was hosted by the Genetics and Genomics Division, Cohen Children's Medical Center/Northwell Health at 2000 Marcus Avenue, New Hyde Park, NY 11042. Dr. Ian Krantz and Kelsey Crocker, MS, CGC offered welcoming remarks and introduced speakers. This day was filled with discussions surrounding ongoing research since the discovery of ReNU syndrome, lightning round talks, and collaborative workshops, with the purpose of establishing a roadmap for future translational research efforts. Key themes emerging from this conference included ending the diagnostic odyssey, clinical presentation and care, family experience, genetic mechanisms, clinical trial readiness, and therapeutic development. Here, we provide a broad overview of the conference and highlight the

key presentations and main priorities for families, researchers, and clinicians moving forward.

## 2 | Discovery

Two research groups independently discovered ReNU syndrome and published peer-reviewed articles describing the condition in mid 2024. Dr. Ernest Turro and Dr. Daniel Greene spoke on behalf of the group from Mount Sinai School of Medicine, while Yuyang Chen and Dr. Nicola (Nicky) Whiffin presented on behalf of the group from the University of Oxford led global collaboration.

Dr. Ernest Turro (Mount Sinai) shared how technological advancements, wider interest in rare conditions, and genome sequencing were important steps in his team's discovery of ReNU syndrome. The National Institute for Health Research (NIHR) and funding for Genomics England and the 100,000 Genomes Project specifically played a critical role, providing an extensive dataset for the discovery of novel genetic causes of rare disease. Dr. Daniel Greene (Mount Sinai) further demonstrated how the Mount Sinai team developed computational and statistical methods, including BeviMed (Greene et al. 2017), a method to test for genetic association between genes and rare diseases, that led to the discovery that *RNU4-2* variants cause the neurodevelopmental condition known as ReNU syndrome. Dr. Greene highlighted that ReNU syndrome is one of the most common monogenic causes of neurodevelopmental conditions, accounting for almost 0.4% of cases in the Genomic Medicine Service managed by Genomics England (Greene et al. 2024).

Yuyang Chen (University of Oxford) shared how the Genomics England dataset was a necessary tool in his research team's discovery of ReNU syndrome. Upon analysis of individuals with neurodevelopmental disorders (NDD), his team identified an 18 base pair region in *RNU4-2* with highly recurrent variants not seen in the general population. These variants map to structural elements in the U4/U6 snRNA duplex, including the T-loop and stem III critical regions. Individuals with ReNU syndrome have significantly more splicing events compared to healthy individuals, suggesting an impaired splicing function (Chen et al. 2024). Dr. Nicky Whiffin (University of Oxford) further summarized what is currently known about ReNU syndrome and efforts to spread awareness, advocate, and develop information resources. She finished this session by looking forward, outlining ongoing collaborative research efforts to tackle key unanswered questions about ReNU syndrome and support diagnosis.

## 3 | Establishing ReNU Syndrome United

When the discovery of ReNU syndrome became public, families finally had an overarching cause for the challenges their children faced for years. For many, it marked the end of an isolating years or decades-long diagnostic odyssey, and the beginning of something powerful: community and renewed hope.

In April 2024, soon after receiving the *RNU4-2* diagnosis, parents created the “*RNU4-2*/ReNU Syndrome Family Connect” Facebook group. Families began joining from around the world

within days. As of mid-2025, the group has close to 500 members globally.

The group gave families a place to ask questions, share experiences, and recognize themselves in one another's stories. Beyond medical descriptions, parents bonded over similar behavioral anecdotes such as children clapping along to music or enjoying water play.

Shortly after the online group formed, Heather Margrill created [renusyndrome.org](https://renusyndrome.org) as a central hub for education and connection, because at the time there was no centralized website for this newly discovered disorder. The site offers access to scientific publications, family stories, advocacy tools, and global community updates. It has quickly become the primary resource for newly diagnosed families as well as medical professionals.

Building off the online community, a group of six family members of ReNU syndrome patients, Jessica Margrill, Lindsay Pearce, Lauren Schwarze, Heather Margrill, Heather Kainz, and Dr. Kim Salazar, came together and incorporated as a nonprofit in October 2024. Thus, officially starting the patient advocacy organization ReNU Syndrome United, dedicated to building community, raising awareness, and advancing research for the ReNU community.

ReNU Syndrome United has grown rapidly since incorporating. Prioritizing a few areas has aided in this rapid growth:

- Establishing a strong network of global scientific advisors and supporters;
- Establishing a patient registry to collect data and advance diagnostics;
- Bringing families and scientists together early on via virtual Q&A panels and this inaugural conference;
- Learning from those who have walked this path previously, including other patient advocacy organizations and rare disease umbrella organizations;
- Leveraging top-tier media coverage and opportunities; and
- Investing time and expertise to build a sustainable organization.

As ReNU Syndrome United continues to lead the patient advocacy journey on behalf of the ReNU syndrome community, the organization used this inaugural conference and scientific symposium to help develop a roadmap for advancing research on ReNU syndrome. Overall, the organization is focused on increasing collaboration, advocating for better diagnostics, and advancing progress on future treatment options.

## 4 | Ending the Diagnostic Odyssey

Another key area of discussion was ending the diagnostic odyssey for ReNU syndrome led by Dr. Seth Berger (Ambry Genetics), which remains a significant challenge due to testing accessibility issues. In the United States, for example, genome sequencing (WGS) has variable payor coverage and is currently required for

diagnosis. The integration of WGS into clinical practice is still relatively recent, resulting in many individuals living undiagnosed with ReNU syndrome despite experiencing symptoms. It is possible for individuals to have previously undergone negative exome sequencing and yet remain undiagnosed due to the need for comprehensive WGS. Although WGS has significantly accelerated novel gene discovery and breakthrough diagnoses within the rare disease community, access remains limited in many parts of the world. According to the international map published on the ReNU Syndrome website, certain countries have better access to WGS than others, contributing to disparities in timely diagnoses (Figure 1). To address these inequities, possible solutions include building *RNU4-2* specific testing and/or capture kits that include *RNU4-2* on exome. While it's possible some labs may consider implementing these solutions, other labs will move to a genome backbone. Discoveries such as ReNU syndrome will drive an increase in the utilization of genome sequencing in the diagnosis of unsolved neurodevelopmental conditions in the coming years. Addressing these testing accessibility challenges is crucial to ensure that individuals globally can benefit from advancements in genomic diagnostics. Further efforts are needed to increase the availability and affordability of WGS in clinical settings worldwide.

Additionally, Dr. Micheil Innes (University of Calgary) highlighted the Canadian landscape of ReNU syndrome with multiple new diagnoses following re-analysis of WGS data. He and others emphasized that the key priorities of the clinical, laboratory, and research teams to expand ReNU diagnoses across the world should include the following: promoting awareness to facilitate genetics referrals and access to WGS, development of in-house targeted assays, enhancement of WGS pipelines to identify known and emerging non-coding small nuclear RNA genes, development of epigenatures, enhancement of facial recognition platforms, establishing large research programs, and inspiring the next generation of clinicians and scientists to work on ReNU syndrome.

## 5 | Clinical Features

Several clinicians and researchers provided a deep clinical characterization of ReNU syndrome patients to date, including developmental delays, intellectual disability, seizure activity, a distinctive facial appearance, ophthalmological features, and musculoskeletal features.

Of note, intrauterine growth restriction can be the first sign of ReNU syndrome, especially in the presence of brain manifestations. ReNU syndrome should also be considered as a differential when there is a clinical suspicion of Angelman or Rett syndrome (Valenzuela et al. 2024). Dr. Scott Perry, MD (Cook Children's Medical Center) emphasized the need to determine the relationship between epilepsy and comorbidities in ReNU patients, and acknowledged no universally known best treatment for seizures in ReNU patients to date. Individuals with ReNU syndrome also often present with bone-related concerns, including mild joint hypermobility, short 3rd and 4th metacarpals, scoliosis, pectus carinatum, osteoporosis, fractures, and hip dysplasia. Dr. Irene Valenzuela Palafoll (Hospital Vall d'Hebron) suggested a low threshold to suspect

RNU4-2 International Map



**FIGURE 1** | International map of *RNU4-2* variants as of January 7, 2026.

and diagnose fractures in patients with ReNU Syndrome, recommending clinical exams to evaluate for skeletal anomalies, a baseline DEXA at 3 years old, and routine screening for osteoporosis (Peñañiel-Sam et al. 2025). Given the distinctive facial features of ReNU syndrome, Dr. Annabelle Arlt (University of Bonn) further suggested GestaltMatcher, an artificial tool for matching facial dysmorphisms with established genetic cohorts through machine learning algorithms, as a useful tool for facilitating diagnostic strategies. To wrap up this topic, a parent also shed light on an understudied symptom of interest for the ReNU community, specifically frequent infections and immune system vulnerability and called for further research related to immune system health for this population.

## 6 | Family Voices

The event included a heartfelt contribution from the siblings of individuals with ReNU syndrome, who shared a story that beautifully illustrated their experiences. This was followed by a panel of courageous parents who shared their personal journeys with ReNU syndrome, offering an intimate glimpse into both the profound joys and unique challenges they encounter as caretakers. A common theme emerged around the joy their children bring through their love for water, hugs, and clapping, which often light up the lives of those around them. However, there were also shared concerns regarding societal perceptions, with parents finding solace when their children are warmly received. The challenges of managing behavioral issues, like hitting and scratching, and frustrations with communication, particularly with caregivers, were candidly discussed. Parents also stressed the importance of avoiding overstimulation and expressed appreciation for the grace of strangers. Through these intimate and honest accounts, both siblings and parents cultivated a deeper understanding of ReNU syndrome and empathy within the community.

## 7 | Evolving ReNU Research

Mirroring the progression of the patient community, the ReNU research community is actively growing and broadening its exploration of various research domains. Dr. Nicky Whiffin (University of Oxford), Dr. Ernest Turro (Mount Sinai), and Dr. Mafalda Barbosa (Mount Sinai) provided insights into some of their ongoing research attempting to understand variation across *RNU4-2* and further characterize RNU-opathies.

Dr. Nicky Whiffin presented a saturation genome editing (SGE) assay based on cell survival. In this effort, single nucleotide variants across *RNU4-2* were introduced to determine which are likely to be pathogenic. She outlined how SGE outperformed computational predictions by distinguishing ReNU syndrome variants from those found in unaffected individuals and correlating them with phenotypic severity. This demonstrated the wider role of *RNU4-2* in neurodevelopmental disorders, including recessive RNU-related disorders and other dominant RNU-related disorders (with the suggested umbrella term of “RNU-opathies”). These SGE scores are freely available as a tool in supporting clinical variant interpretation, and they reveal distinct mechanisms that will help inform therapeutic approaches for the ReNU syndrome community (De Jonghe et al. 2025). Dr. Whiffin further discussed the state of animal models created to better understand the mechanism of action and to explore therapeutic approaches with a mouse model created in collaboration with Jackson Laboratories.

Dr. Ernest Turro presented his team's statistical work to discover novel genetic causes of neurodevelopmental disorders (NDDs). Their analysis demonstrated that variants in the closely related snRNA genes *RNU4-2* and *RNU2-2* cause two of the most prevalent NDDs with low genetic heterogeneity (Greene et al. 2024, Greene et al. 2025). Dr. Turro explained that, in collaboration with Dr. Kathleen Freson (KU Leuven), the team has

introduced pathogenic *RNU4-2* variants into induced pluripotent stem cells (iPSCs) and differentiated them into neuronal stem cells and neurons, to study the molecular mechanisms of ReNU syndrome in disease-relevant cell types. Dr. Turro also described his collaboration with Dr. Bekim Sadikovic of EpiSign showing that methylation variant pathogenicity scores discriminate ReNU syndrome from non-ReNU NDDs with high specificity. This preliminary work supports the establishment of a diagnostic marker to assist with variant interpretation. Finally, Drs. Turro and Barbosa presented the INDEED study at Mount Sinai which uses morphological pattern recognition algorithms (Face2Gene) coupled with genetic testing to identify patients currently without a genetic diagnosis. The study will characterize the natural history of ReNU and related syndromes, develop clinical management guidelines and acquire biospecimens for mechanistic studies and future trials.

Additional research focused on variant exploration and interpretation, clinical and functional studies to better identify and understand ReNU syndrome, and the importance of creating a centralized biorepository.

Dr. Hector Rodrigo Mendez described his multiomic approaches to identify and characterize pathogenic variants in small nuclear RNAs (snRNAs), including *RNU4-2* and *RNU4ATAC*, at the Stanford Center for Undiagnosed Diseases and the GREGOR Stanford Site. His work emphasized the importance of noncoding variants and transcriptomic-wide approaches in diagnosing and understanding spliceopathies (Arriaga et al. 2025; De Jonghe et al. 2025). Dr. Stefan Barakat (Erasmus MC University Medical Center) presented clinical studies, functional genomics approaches, and computational methods aimed at efficient reanalysis of previously unsolved ReNU syndrome cases, disease modeling of ReNU syndrome using a zebrafish model, and the potential of artificial intelligence models to identify noncoding causes of conditions (Schot et al. 2024; Deng et al. 2025). He spoke to the importance of genome sequencing and the characterization of non-coding genes in the diagnosis of other unsolved individuals in upcoming years. Dr. Michael Sheldon (SAMPLED) concluded this topic with a discussion on the importance of integrated biobanking and analytical services for rare disease research, with key takeaways of creating a central repository for consistent sample collection and cell services, as well as easy to use web-based interfaces to ensure ease of sample access for ReNU research moving forward.

## 8 | Clinical Trial Readiness

One recurrent theme of the conference was the need for quality natural history data in the path to clinical trial readiness. Dr. Maya Chopra (Boston Children's Hospital) outlined the GENE TARGET framework, which provides a systematic way to evaluate monogenic neurodevelopmental disorders for suitability for therapeutic development (Chopra et al. 2022). While *RNU4-2* is an excellent target due to its prevalence, potential for early diagnosis, recurrent variant, favorable risk-benefit ratio, and engaged community, gaps persist in several domains—most notably in establishing validated endpoints and gathering natural history data. Endpoints for neurocognition can be difficult to establish, and it is important to ensure they are meaningful to

families. She concluded with an emphasis on the importance of family and advocate voices to inform the prioritization of treatment goals for future clinical trials.

## 9 | Therapeutics

Therapeutic investigation and innovation were a major theme throughout the conference. Dr. Justin Ichida (BioMarin Pharmaceutical Inc.) defined key areas to address for therapeutic development and pediatric indications, including understanding the genetics of ReNU syndrome, establishing a therapeutic target(s), access to natural history data, and remaining commercially competitive. Different symptoms are also important to investigate and address from a therapeutics standpoint. Jeff Milton (La Jolla Labs and St. Jude's) further emphasized the importance of understanding the mechanism of action in ReNU syndrome to develop an effective therapeutic and shed light on the expensive development and extensive therapeutic discovery journey that investigators will face.

Different potential therapeutic strategies for ReNU syndrome were then unpacked by current researchers. Dr. Stanley Crooke and Dr. Konstantina Skourti-Stathaki (n-Lorem Foundation) dove into ASO therapy in relation to ReNU syndrome, with the goal of silencing the *RNU4-2* variants with an allele-specific ASO. They presented preclinical research suggesting that *RNU4-2* variants may be a good candidate for ASO therapy. The n-Lorem team is continuing to explore the mechanism of action, moving forward with rat tolerability studies, and is cautiously optimistic about the path ahead. Stefan Lukianov (Salve Therapeutics) proposed to computationally design a gene therapy for ReNU Syndrome using a machine learning-based tool called VirCAD (virus computer-aided design) that has been developed to study protein-protein interactions. This computational design platform enables the identification of specific neuronal surface receptors and guides the rational modification of the adeno-associated virus 9 (AAV9) capsid to optimize binding affinity, thereby enhancing the precision of gene delivery of the *RNU4-2* genetic payload. Dr. John Rinn (University of Colorado Boulder) highlighted some potential therapeutic strategies, including antisense oligonucleotides (ASO), CRISPR technology, and/or RNA replacement therapies. He further presented on the temporal dynamics of long noncoding RNA (lncRNA) in mediated epigenetic and transcriptional regulation of NDDs, suggesting lncRNA U4 may also be an important avenue to explore in developing a treatment for ReNU syndrome (Much et al. 2024).

The following works were also presented in a poster presentation format:

- Jillian O'Toole, MS, CGC (Boston Children's Hospital) presented her research titled: "Accelerating Clinical Trial Readiness for ReNU Syndrome."
- Brooke Friedman, BA, neuroscience PhD candidate (Icahn School of Medicine at Mount Sinai) and New York State Ambassador (National Organization for Rare Disorders) presented her research titled: "Leveraging Long-Read RNA-seq to Define TDP-43 Associated Isoform Changes in Alzheimer's Disease."

- Konstantina Skourti-Stathaki, PhD (n-Lorem) presented her research titled: “ReNU Syndrome Therapeutic Research Efforts.”
- Robert Rigobello, MS, CGC (Baylor Genetics) presented his research titled: “Utility of Whole Genome Sequencing to Detect *RNU4-2* Variants Associated with ReNU Syndrome.”

## 10 | Advocacy

Panelists Lindsay Pearse (ReNU Syndrome United Board Member), Sarah Warwick, and Rachel Johnson discussed the endeavors of the past year to explore media opportunities, share lived experiences, and partnerships they've built within the research community. They also highlighted the ReNU Syndrome map created by Heather Margrill (Figure 1), announced the ReNU Awareness Days (February 4th and April 2nd), ReNU Awareness Month (April), fundraising initiatives to raise money for research, and the very first ReNU Unique guide for distribution to families and clinicians (Unique, N. Whiffin et al. 2025). The ongoing collaboration with media and research partners is viewed as crucial for driving forward both scientific understanding and social acceptance, ultimately aiming for a future where ReNU syndrome is more widely recognized and understood.

## 11 | Summary

The inaugural ReNU Syndrome conference marked a significant milestone in the study and understanding of this newly discovered disorder. It is truly remarkable that this non-coding, previously uncharacterized, 145 base pair gene is estimated to explain ~0.4% of all individuals with a neurodevelopmental disorder (Chen et al. 2024). This finding underscores the importance of non-coding genes in rare disease, and it is surely not the last to make an impact in this new era of genome sequencing. In the time span of a little over a year, *RNU4-2* was discovered, a patient advocacy group was formed, and an international collaborative conference was planned and executed. This conference, which brought together patients, families, scientists, physicians, and researchers, illustrated the impressive momentum and collaborative spirit of this community dedicated to advancing therapeutics for non-coding genes.

The workshop at the end of the conference, led by Dr. Christina SanInocencio (Rare Epilepsy Network), facilitated lively discussions surrounding main priorities for the clinical and research advancement of ReNU syndrome. The recommendations are as follows:

1. *Natural History Survey*: Initiate and fund a globally accessible survey and establish best practices for data collection in collaboration with advisors
2. *Mechanism of Action*: Focus on the development and expansion of animal models to understand disease mechanisms.
3. *Consensus Guidelines*: Regularly update clinical guidelines and resources such as GeneReviews, supported by a dedicated clinical working group.

4. *Biomarkers*: Develop robust biomarkers by funding studies across diverse populations.
5. *Registry/Biobank*: Establish and fund a centralized global registry to facilitate sharing of iPSC lines, organoids, and cell lines.
6. *Data Sharing*: Foster partnerships to encourage data sharing.
7. *Awareness and Access*: Advocate for broader inclusion of ReNU syndrome in genetic testing panels, enhance awareness to facilitate genetic referrals, and inspire future generations of clinicians and scientists.

The insights and collaborations from this conference outline a comprehensive strategy to tackle ReNU syndrome and *renew hope* for this community. The collective efforts set a promising foundation for the development of natural history studies and therapeutic interventions and reinforced the importance of continued research and collaborative partnerships in advancing our understanding and treatment of rare genetic conditions. This extraordinary historic event brought together families, scientists, clinicians, and industry leaders in the newly formed ReNU syndrome community and truly served to ReNU Hope for a brighter future for all.

### Author Contributions

**Kelsey Crocker**: writing – original draft, writing – review and editing. **Jillian O'Toole**: writing – original draft. **Lindsay Pearse**: writing – original draft. **Jessica Margrill**: writing – original draft. **Asbaa Khan**: writing – original draft. **Maya Chopra**: writing – review and editing, conceptualization. **A. Micheil Innes**: conceptualization, supervision. **Heather Kainz**: writing – review and editing. **Heather Margrill**: writing – review and editing. **Kim Salazar**: writing – review and editing. **Lauren Schwarze**: writing – review and editing. **Ian D. Krantz**: writing – review and editing, supervision. All authors reviewed and approved the final manuscript.

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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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